Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/22/2011
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Project Title:	Gradient Driven Fluctuations-Phase 2		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Santa Barbara	**State:**	CA
Zip Code:	93106-9530	**Congressional District:**	23
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	03/01/2008	**End Date:**	08/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	Foton		
Flight Assignment:	Foton M3 NOTE: New end date is 8/31/2012 per NSSC information (Ed., 6/20/2012) NOTE: New end date is 2/28/2012 per NSSC information (Ed., 5/31/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal Text:			

Task Description:	GRADFLEX, a co-operative effort between NASA, ESA, the University of Milan and the University of California at Santa Barbara (UCSB) was flown aboard the Foton-M3 satellite mission in September 2007. The apparatus applied various thermal gradients to both a sample of a single-component fluid (carbon disulphide) (SINGLE) and a 2-component mixture (polystyrene in toluene) (MIXTURE). Thermal fluctuations under these conditions are greatly enhanced both in amplitude and spatial extent. This effect occurs on Earth, but is limited by gravity, while in micro-gravity, the range of the fluctuations is predicted to be limited only by the sample dimension. A sensitive shadowgraph method capable of carrying out both static and dynamic light scattering at scattering angles as small as one thousandth of a degree was developed and used for this experiment. This method is now beginning to be applied in other areas, such as colloidal physics. The flight returned approximately 60 Gb of data images, while ground-based runs, completed using the same apparatus after flight, added another 400 Gb of data. This grant requests support for the sorting, decompression and analysis of this massive data set. Because the instrument behaved in an unexpected way when gradients were applied during flight (but not after it was safely back on Earth!), the question of how to analyze the flight data has no obvious a priori answer. Initial attempts have shown that unexplained spectral power is present for low wave vector, and this spectral power appears in the form of a narrow band, very low frequency, noise process. The relative contribution of this noise source varies with position in the cell, and from run to run. It is stronger with increasing gradient. The fact that it has a temporal power spectrum that differs markedly from that of the gradient-driven fluctuations offers the possibility that it can be separated via spectral analysis. Another complication that must be addressed is the role played by optical imperfections on the analysis of shadowgraph data. This unfortunate question has arisen because the sapphire window used in flight for SINGLE was not of the highest quality. It appears perfectly possible to expand the theory of the shadowgraph to include static imperfections, and this will be attempted before full data analysis is undertaken. There appear to be only relatively minor problems with both the static and dynamic data obtained on Earth after flight, and they should not be insurmountable. If the spectral method of analyzing the flight data is successful, the same method will be used for the ground-based data in order to be consistent. The questions to be answered from the analysis are: 1. Does the predicted spectral shape for micro-gravity describe the flight data well? 2. Is the mean-squared amplitude of the fluctuations S(q) on Earth, compatible with the predicted value, in absolute terms? If this question can be answered, even to within 10% accuracy, it will be an excellent test of theory. It would also represent an unprecedented technical achievement in the field of light scattering. 3. Does the removal of gravity, enhance the mean-squared amplitude of the fluctuations and alter the shape of S(q) to the extent and in the manner predicted? 4. Do the dynamics measured on Earth, show the same sort of minor differences from what is predicted as did the data obtained using the UCSB prototype instrument and which have already been published (Physical Review Letters, 100, 234502 (2008))?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Throughout the course of this work it has been necessary to constantly improve the experimental method of choice, namely the quantitative shadowgraph. This development has now reached the stage where it is clear that this method, if handled properly, represents a viable method of carrying out both dynamic and static light scattering at scattering angles so small as to defy the imagination of anyone who has worked in the field. Although it is likely to take some time for the knowledge we have gained to be disseminated, there is no reason this method cannot be added to the scattering arsenal. Furthermore, it can also be applied to X-rays, enabling one to extend the lower limit of X-ray scattering as well. The consequences of putting a new technique into the hands of researchers is difficult to asses in advance. Basically, however, this method will enable quantitative measurements of various systems having size ranges that are too large for existing techniques. Obvious examples are the crystallization process in colloidal systems, gelation processes, and aggregation and flocculation. Because scattering provides excellent statistical averages, the method should prove quite complementary to visualization techniques such as microscopy.
Task Progress:	The science team in Milan has completed the analysis of the data obtained from a mixture of low molecular weight polystyrene dissolved in toluene (Mixture experiment), and the results have been published in Nature Communications. The paper (Nature Commun., http://dx.doi.org/ (2011)) can be accessed online. This publication includes a "movie" showing the development of extremely large amplitude, very long wavelength, extremely slow, fluctuations in this sample both on Earth and during flight, after application of a temperature difference across the 1 mm thick sample. To access the movie choose "supplementary information." As the eye discerns immediately, the absence of gravity greatly enhances the fluctuations. More precise analysis of the data shows it to be consistent with theory, although no absolute amplitude comparison has yet been made. The science team at UCSB was faced with a more difficult task. Data obtained during flight contained excess noise of an unknown origin, that was present at low frequencies and long wavelengths. A promising method had been devised for extracting the gradient driven fluctuations from this data in the previous 8 months, and this method has now been applied to all of the flight data, enabling us to extract the amplitude of the fluctuations as a function of their wavelength. In addition, it has proved possible to calibrate the instrument in absolute terms, so that the data can be compared directly to the theory, without any adjustable parameters. In other words, the theory has no place to hide! The results of this analysis have just been published in Physical Review Letters (Phys. Rev. Lett. Vol. 106, p. 244502 (2011). The results are a resounding victory for the theorists. Their predictions agree with our data to withing a few percent on Earth and seem to be about 15% too large in micro-gravity, however given the mysterious noise source, we cannot say this discrepancy is serious.
Bibliography Type:	Description: (Last Updated: 02/27/2013)

Articles in Peer-reviewed Journals	Vailati A, Cerbino R, Mazzoni S, Takacs CJ, Cannell DS, Giglio M. "Fractal fronts of diffusion in microgravity." Nature Communications., 2011;2:290. http://dx.doi.org/10.1038/ncomms1290 ; PMID: 21505446 , Apr-2011
Articles in Peer-reviewed Journals	Takacs CJ, Vailati A, Cerbino R, Mazzoni S, Giglio M, Cannell DS. "Thermal fluctuations in a layer of liquid CS2 subjected to temperature gradients with and without the influence of gravity." Physical Review Letters. 2011 Jun 17;106(24):244502. http://dx.doi.org/10.1103/PhysRevLett.106.244502 , Jun-2011